THANMAYEE BOYAPATI

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EDUCATION

University of Minnesota -Twin Cities December 2025

Major: Computer Science B.S & Data Science B.S | **Minor:** Statistics

SKILLS

Languages: Python, SQL, Java, JavaScript, C, R,

ML/AI: PyTorch, Scikit-learn, Hugging Face, Sentence Transformers

GenAI: LLM Fine-Tuning, RAG, Embeddings, Prompt Engineering, Evaluation Metrics, LangChain

Data & Systems: Pandas, NumPy, Docker, AWS, GCP, MongoDB, REST APIs, Tableau

RELEVANT EXPERIENCE

Group Lens Research Lab | Minneapolis, MN August 2024 - Present

ML Research Assistant (Personalization & NLP)

- Designed a hybrid intent-alignment pipeline that processed 1000+ news headlines, utilizing RoBERTa and Sentence Transformers to quantify semantic alignment with 90%+ accuracy against human-labeled benchmarks
- Accelerated failure mode identification by developing diagnostic analytics and visualization workflows that surfaced click-through-rate (CTR) trends and locality-category shifts across headline generation strategies
- Led 50+ controlled experiments by testing different prompts and topics to evaluate model performance; analyzed results to reduce inconsistent outputs and provided the core data for a forthcoming paper on News Recommendation Systems

Mayo Clinic | Rochester, MN June 2025 - December 2025

AI Engineer Intern - Biostatistician Department

- Optimized clinical query accuracy by 20% by architecting an end-to-end genomics RAG system that integrated OCR, entity extraction, and embedding-based retrieval to ground LLM responses in domain-specific data
- Reduced retrieval latency by 350ms by designing a high-performance retrieval layer using MongoDB and LangChain to index and query structured clinical entities and unstructured document chunks during LLM inference
- Improved contextual relevance of a domain-adapted Llama-3 model by fine-tuning with Sentence Transformer similarity metrics and iteratively refining context selection strategies

PROJECTS

Brain-Based Prediction of tDCS Treatment Response | *Python, Scikit-learn, Causal Inference*

- Constructed a comparative analysis of high-dimensional brain connectivity representations to predict individualized treatment responses using Causal Inference models like T-Learner, X-Learner, and Causal Forests
- Optimized model generalization on small-sample clinical neuroimaging data by implementing advanced dimensionality reduction, achieving a 5% improvement in predictive accuracy over baseline models

LLM-Powered Financial Analytics Platform | *React, Node.js, MongoDB, OpenAI API*

- Architected a full-stack financial analytics platform using React and Node.js with a MongoDB persistence layer, enabling scalable storage and retrieval of unstructured transaction records
- Integrated an LLM-driven data pipeline to automate the categorization of financial activity, transforming raw transaction data into personalized spending insights and reducing manual entry for users

RegEx-Based Medical Data Parser | *Python, Tesseract OCR, RegEx*

- Developed an OCR-driven extraction pipeline that transformed unstructured medical PDFs into structured JSON formats, specializing in the recovery of visa authorizations, USMLE outcomes, and ECFMG certifications
- Engineered complex Regular Expression (RegEx) patterns to parse high-variability document layouts, ensuring high-fidelity data extraction for downstream clinical processing

VPN-Based Remote Robotics Interface | *Linux, Networking, Site-to-Site VPN*

- Deployed a secure Site-to-Site VPN infrastructure to facilitate real-time, low-latency communication between autonomous robots and remote driver stations, supporting critical telemetry for hardware testing
- Sustained network reliability for a distributed team of engineers in Poland and the U.S., ensuring 98% uptime and secure resource access during international robotics competitions

HONORS & LEADERSHIP

- NCWIT Aspirations in Computing – ***State Winner***
- *CSCI 5525 Machine Learning – Teaching Assistant*
- College of Science & Engineering – *Dean's List*
- South Indian Student Association (SISA) – *Co-Founder*
- NCWIT – National Center for Women & IT – *Campus Representative*
- College of Science & Engineering – *CSE Ambassador*